Growth-based school accountability rules and grade retention practices

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Will Labadie

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Abstract

Do accountability rules affect public school retention practices? Using a simple model of grade retention, I show that an administrator will retain students differently depending on the accountability ratings criteria he seeks to maximize. I show that an administrator whose school is rated based on student proficiency (based on standardized exam scores) has a strategic incentive to retain borderline students, while an administrator whose school is rated based on both proficiency and individual student test score growth has incentive to retain only the lowest-scoring students. I further show that this effect is most pronounced in the final grade offered by a given school, when promotion of a student ensures her removal from the school's pool of test-takers. I test the predictions of my framework using a novel dataset of school-grade level retention rates for seven states and an event study design. I find that on average, exposure to growth-based accountability criteria reduces terminal-grade retention rates by about 29%. The effect appears to intensify with time, reaching 35-42% after 2-4 years of exposure.

1 Introduction

Each year, public school administrators endeavor to guide their schools to satisfactory ratings as defined by their state's adopted accountability system. It is well-established that administrators value maximizing their schools' ratings, and that they respond to the specific criteria by which their schools are rated (Rockoff and Turner, 2010; Chakrabarti, 2013). Accountability systems generally do not include targeted goals around student retention, but an administrator's retention practices could have significant impacts on her school's overall rating. To the extent that administrators decide whether or not to retain a student with an eye towards optimizing their schools' ratings, different ratings criteria could lead to different retention decisions.[1]

[1] "Retention" is used to refer to various objects in education; here' I use it to refer to *grade* retention, which is sometimes referred to as the practice of holding a student back, or non-promotion, or grade repetition. Retention can refer to student non-attrition, particularly in higher education literature. A large literature examining the determinants of teacher labor market decisions explores teacher retention, the ability of schools to keep their teachers on-staff. Throughout this paper, I use the term "retention" exclusively to refer to grade retention.

In this study, I evaluate the impact of a common accountability measure, year-to-year growth in student standardized test scores (hereafter, I will refer to an accountability system including such a measure as "growth-based accountability"), on public school retention policy. Retention has been shown to have profound effects on students in the short and long term, but the school-side determinants of retention are not well-understood (Schwerdt, West and Winters, 2017; Eren, Lovenheim and Mocan, 2018). This paper sheds light on how administrator incentives could affect the retention decision.

I find that the effect of growth-based accountability criteria on retention is more pronounced in the final grade offered by a school. This result is consistent with the predictions of my statistical framework. In particular, I find that on average, exposure to growth-based accountability criteria reduces terminal-grade retention rates by about 29%. The effect appears to intensify with time, reaching 35-42% after 2-4 years of exposure. This result may be a consequence of strategic behavior on the part of school administrators, as the decision to retain or promote a student in the terminal grade results in either the inclusion of the student in the pool of students whose tests scores contribute to the school's ratings for an additional year, or his removal from that pool. However, my empirical analysis does not allow me to rule out other explanations for growth-based accountability causing reduced retention and differential effects in terminal grades.

On the whole, my results may suggest that fewer students are strategically retained on average under growth-based accountability systems. To the extent that passing rate-based systems encouraged retention of students with the goal of maximizing passing rates, this is a good thing. Prior research on retention has found substantial negative effects on students in the long term (Eren, Lovenheim and Mocan, 2018; Brodaty, Gary-Bobo and Prieto, 2013). This is especially true for older retained students (Jacob and Lefgren, 2009). It is possible that the finding of decreased retention is driven by more targeted retention, since schools' ratings under growth-based accountability systems would be negatively affected by retaining students that would be developmentally harmed by retention. This also would be a positive result, since the weight of evidence seems to suggest that retention is most beneficial when used as a highly individualized intervention (Jacob and Lefgren, 2004; Fruehwirth, Navarro and Takahashi, 2016).

To study the effects of growth-based accountability systems on retention practices, I develop a statistical framework in which students' test scores progress over time and administrators set score thresholds for each grade; students that score below the relevant score threshold are retained, while those that score above the threshold are promoted. This framework predicts different score threshold patterns for maximizing school-wide passing rate and sufficient score growth rate, consistently predicting lower retention in the final grade offered by a school. School administrators' incentives change with the accountability rules their schools are subject to. Through retention,

an administrator has a degree of control over the difficulty of test faced by a given student. If the administrator expects that a grade-4 student is unlikely to pass the grade 5 exam, but likely to pass the grade 4 exam, retaining that student could have a non-negligible effect on his school's accountability rating. Under an accountability system based only on school passing rate, strategic retention of this sort could have an especially clear effect on the school's accountability score. However, under a system that rates schools on both passing rate and within-student score growth, the effect could be less clear. In the framework I develop, students with different ability levels affect a school's passing rate and sufficient score growth rate differently upon retention. In the model, the higher the student's score, the more likely she is to pass the subsequent year's exam if retained. However, students with higher latent ability in the model are less likely to exhibit sufficient score growth if retained than those with lower latent ability. The model predicts that the change in retention practices caused by the adoption of a growth-based accountability system will be especially pronounced in the final grade offered by a school. Students that would increase the school's passing rate if retained are likely to have relatively high latent ability in the model, and are less likely to exhibit sufficient score growth. In the final grade offered by the school, these students can be removed from the pool of students whose scores contribute to the school's rating if promoted. In earlier grades, such students are likely to exhibit insufficient score growth whether promoted or retained, and will negatively affect the school's sufficient score growth rate equally.

In this study, I show that administrators retain students differently depending on the accountability criteria they face. Almond, Lee and Schwartz (2016) show that public school administrators do exercise influence over the retention decision, though this study is the first to show that they do so strategically. A large literature exploring the unintended consequences of accountability systems has established already that administrators value satisfactory ratings (Rockoff and Turner, 2010). In this paper, I establish retention as a tool that administrators are willing to use in pursuit of better ratings for their schools. This paper contributes to a large literature exploring the tools administrators use to maximize their schools' ratings, including highly individualized tools (Figlio and Winicki, 2005; Reback, 2008). I show that administrators alter the body of students that contribute to their schools' ratings. Cullen and Reback (2006) find evidence of schools altering their test-taking pool by strategically exempting certain students from taking state exams to improve their ratings.

The rest of the paper is structured as follows. Section 2 will elaborate on school accountability and the incentives of administrators. Section 3 lays out my statistical framework. Section 4 discusses the data I use to test the predictions of the framework. Section 5 details the empirical strategies I use and the results of the analyses. Section 6 concludes.

2 Background

Since the enactment of the No Child Left Behind (NCLB) Act of 2001, states have been required to evaluate public schools based on student performance on standardized math and English Language Arts (ELA) exams. Under NCLB, state education agencies were required to evaluate schools based on the rate at which students passed its standardized exam, and on the rate at which various subpopulations of students at the school passed the exam. However, NCLB allowed state education agencies a substantial degree of flexibility in designing their accountability systems; each agency could choose its own standardized exam to administer, and define what level of performance would be considered adequate, and accountability systems could and did evaluate schools on criteria beyond the minimum competency requirement. A common criteria, especially among states that had accountability systems in place prior to the enactment of NCLB, was year-to-year growth in individual students' scores.

The Every Student Succeeds Act (ESSA) of 2015 requires state education agencies to develop accountability systems that include multiple evaluation criteria, which must include "students' performance on the statewide assessment, high school graduation rates, and English language proficiency" as well as at least one additional measure of school quality that is left up to the individual agencies (Alliance for Excellent Education, 2016).

Figure 1 shows the percentage of U.S. states whose accountability systems included a measure of student test score growth from the 2004-05 to 2017-18 school years. In the 2004-05 school year, 9 states evaluated schools partially based on the growth rate of students; by the 2017-18 school year, every state did. While states did not randomly choose whether or not to adopt growth into their accountability systems, the decisions are exogenous to school-level decision-making, and affect the way in which schools are evaluated.

A school administrator's incentives are affected when the measures by which his school is evaluated change. Once student score growth contributes to his school's rating, the administrator must attempt to satisfy the state-defined criterion, along with the other indicators the state education agency considers, in order to maximize his school's overall rating. A deep body of research has shown that authorities at schools use a number of tools to increase their school's overall rating. Reback (2008) finds that Texas students whose scores are relatively important to their school's overall rating perform better than expected, and finds evidence of finely targeted resource allocation and instruction to these "high-leverage" students; Figlio and Winicki (2005) find that schools in Virginia at risk of accountability-based sanctions increase the caloric content of school lunches on test days; Craig, Imberman and Perdue (2013) find that district administrators increase instructional budgets after facing a negative rating shock. While Cullen and Reback (2006) find that Texas schools manipulate the pool of test-takers via exemptions, no study has examined whether or not administrators hold students back strategically as well.

Under a passing rate-based accountability system, a ratings-motivated administrator might hold a student back for a number of reasons. If the student is in grade g and the administrator expects the student to pass the grade g exam in the coming school year, but not the grade $g+1$ exam, he might retain the student to bolster his school's passing rate. This is particularly true if the student is from one of the subpopulations whose passing rates are explicitly valued under his state's accountability system. The administrator might also retain the student if he believes that retention will positively impact the student's development, and have a lasting and positive impact on the student's future scores. Some research has shown that retention has positive short-run effects on student scores, but negative long-run effects on student development (Schwerdt, West and Winters, 2017). There is a great deal of heterogeneity in the effects of retention on students, and these factors are likely to play into administrators' retention decisions. However, a purely ratings-motivated administrator would make them all based on students' current or future probability of passing the standardized exam.

Under an accountability system that includes a measure of student score growth and of passing rate, the incentives to retain change. On one hand, an administrator might be more likely to retain a student if he expects the short-run benefits to be high enough; it would improve his school's likelihood of being rated well on both school ratings criteria in the short term. On the other hand, if retention has persistent negative effects, holding a student back will make it less likely that his school receives high ratings for student score growth in the future.

3 Conceptual framework

Because the overall effect of retaining a student on a school's rating is unclear, I build and simulate a basic one-school model of grade retention. I build an extremely simplified model in which students accumulate skills each year they are in school, which translate into standardized test scores. The administrator of the school observes test scores, and sets promotion thresholds to maximize his schools rating. Students with test scores above the promotion threshold set by the administrator are promoted, while students scoring below the promotion threshold are retained. The administrator's objective in this model is to allocate students across grades to maximize either the percentage of students passing the standardized exam (passing rate) or both the passing rate and the percentage of students exhibiting sufficient test score growth from one year to the next (sufficient growth rate). I compare the retention practices predicted by the model for an administrator whose school rating is based on passing rate alone and an administrator whose school rating is based on both passing rate and sufficient growth rate.

This model abstracts away from numerous important realities which affect the retention decision. Administrators' objective functions in reality include much more than just school ratings, though administrators have been shown to respond robustly to school ratings. The retention decision is, in reality, not the administrator's decision to make alone; parents and teachers are generally consulted in the decision. The reten-

tion decision is also generally unlikely to be made based on standardized test scores alone in reality. The goal of this paper is to evaluate the effect of the incentives created by accountability systems on administrator behavior with respect to retention; this model uses an extreme case which does not reflect reality, in which administrators are purely motivated by school ratings, to examine the mechanisms through which passing rate-based and sufficient score growth-based accountability ratings criteria may affect administrator retention practices.

The model relies on a factor model formulation of student skill accumulation and skill measurement, following the basic structure of Cunha and Heckman (2008). In each period t, students at the school earn a score on the state's standardized exam (s). Assume that there is only one subject exam. Assume that student scores are represented by the following dynamic factor structure:

$$s_{it} = \mu + \alpha(g_{it})\theta_{it} + \epsilon_{it} \tag{1}$$

where i represents the individual student, and θ_{it} is a dynamic factor for each student. $\alpha(g_{it})$ is the factor loading parameter, which translates latent ability θ_{it} into a measurable standardized test score, s_{it}. It is a function of the student's grade, g_{it}, reflecting the fact that students in different grades take tests of different difficulty levels; I assume that $\frac{\partial \alpha(g_{it})}{\partial g_{it}} < 0$, such that students with the same latent ability values will earn lower scores in expectation if they are placed in higher grades. ϵ_{it} represents a random shock to the measured scores, representing the noisiness of test scores in measuring ability.

I assume the following form for students' skills production:

$$\theta_{it} = \gamma_0 + \gamma_1 \theta_{it-1} + \eta_{it} \tag{2}$$

where η_{it} is assumed independent across students and over time for the same students. I assume that students draw some initial value θ_{i0} prior to entering school. For simplicity, I assume that retention does not affect the skills production function. Research suggests that it would be more realistic to include retention in the skills production function, and this is a consideration for future work.

The administrator's objective in this model is to allocate students across grades to maximize either the passing rate of his school or both the passing rate and sufficient growth rate of the school. The school's passing rate in year t is given by

$$\Pi_t = \frac{\sum_{j=1}^{G} \sum_{i=1}^{N} \mathbb{1}(g_{it} = j) \times \mathbb{1}(s_{it} \geq \pi)}{N_t} \tag{3}$$

where G represents the final grade offered by the school and N represents the total number of students. π represents the externally set passing exam score. N_t represents the number of students whose scores count towards the school's rating in year t:

$$N_t = \sum_{j=1}^{G} \sum_{i=1}^{N} \mathbb{1}(g_{it} = j). \tag{4}$$

The school's sufficient growth rate is assumed to be measured as follows:

$$\Lambda_t = \frac{\sum_{j=1}^{G}\sum_{i=1}^{N} \mathbb{1}(g_{it} = j) \times \mathbb{1}(s_{it} - s_{it-1} \geq \lambda)}{N_t} \tag{5}$$

where λ represents the externally set target amount of score growth for a given student, and all other objects are as defined above. In reality, accountability measures of student growth vary across states; however, the percent of students demonstrating a sufficient level of score growth is a central component of all the growth-based accountability measures I analyze in this paper.

The school administrator is able to choose the difficulty of the exam that a student is exposed to in a given year through the retention decision. The administrator can choose to retain a grade g student, and she will take the grade g exam rather than the more difficult grade $g+1$ exam. I assume that administrators make the retention decision by setting promotion thresholds δ_g for each grade. Grade g students that score at or above δ_g are promoted to grade $g+1$, while grade g students that score below δ_g are retained and repeat grade g. That is, administrators control students' grade level g based on the following:

$$g_{it} = g_{it-1} + \mathbb{1}(s_{it-1} \geq \delta_g). \tag{6}$$

I assume that retention is the only tool administrators have to influence their school's scores in this model; they can only affect Π_t and Λ_t through their δ_g choices. δ_g affects student scores, s_{it}, directly through $\alpha(g_{it}(\delta_g))$, but it also affects the population of grades g and $g+1$ through N_t, since g_{it} is a function of δ_g.

I assume that the school administrator has two important constraints. First, he must promote students that pass the standardized exam:

$$\delta_g \leq \pi_g \tag{7}$$

and he cannot retain a student in the same grade two years in a row:

$$g_{it} \neq g_{it-2} \forall i. \tag{8}$$

These constraints are based on common rules adopted by school districts and state education agencies. The first has the additional feature of removing the option of retaining all passing students to maximize the school passing rate, which is appealing for the simulation of this model. I assume that these constraints hold for administrators regardless of the accountability system their schools are subject to.

An administrator of a school under a passing rate-based accountability system has the following optimization problem:

$$\max_{\delta_1,\delta_2,\delta_3} \Pi(\delta_1, \delta_2, \delta_3) \quad s.t. \quad \delta_g \leq \pi_g, \quad g_{it} \neq g_{it-2} \forall i. \tag{9}$$

while an administrator of a school under a passing rate- and sufficient growth-based accountability system has the following optimization problem:

$$\max_{\delta_1,\delta_2,\delta_3} \Pi(\delta_1,\delta_2,\delta_3) + \Lambda(\delta_1,\delta_2,\delta_3) \quad s.t. \quad \delta_g \leq \pi_g, \quad g_{it} \neq g_{it-2} \forall i. \tag{10}$$

The mechanisms driving the administrator's decision are shown graphically in Figure 2. Figure 2a plots the relationship between a student's grade level, his latent ability, and his test score. Because the α function is decreasing in student grade level, the same level of latent ability will translate into lower scores for students in higher grades in expectation. A given student i with latent ability θ_{it} in year t earns a test score equal to $s_{it}(\theta_{it}, \alpha(g_{it} = j))$ if he is in grade j in year t. Figure 2a shows a student whose latent ability grows from θ_{it} in year t to θ_{it+1} in year $t+1$, and the scores he would earn in year $t+1$ if he is retained or promoted. $\bar{s}$ represents the maximum achievable score. If retained, the student will earn a score of $s_{it+1}(\theta_{it+1}, \alpha(g_{it+1} = j))$; if promoted, his score will be substantially lower, at $s_{it+1}(\theta_{it+1}, \alpha(g_{it+1} = j+1))$. If π, the externally-set passing standard, is between $s_{it}(\theta_{it}, \alpha(g_{it} = j))$ and $s_{it+1}(\theta_{it+1}, \alpha(g_{it+1} = j+1))$, the administrator's objective function is expected to be as well-served by promoting the student as by retaining him. In this case, an optimal δ_1 will be set below $s_{it}(\theta_{it}, \alpha(g_{it} = j))$. If π is between $s_{it+1}(\theta_{it+1}, \alpha(g_{it+1} = j+1))$ and $s_{it+1}(\theta_{it+1}, \alpha(g_{it+1} = j))$, however, the administrator would be best-served by retaining the student. If promoted to grade $j+1$, the student would not be expected to pass the grade $j+1$ exam; if retained, the student would be expected to pass the grade j exam.

Figure 2b shows the relationship between a student's grade level, his latent ability, and the rate of change in his scores. For the same values θ_{it} and θ_{it+1} plotted in Figure 2a, Figure 2b plots the partial derivative of s with respect to θ. The shape is determined by the functional form of α, since α is a function of grade level, which is determined by score - a function of θ. If λ, the externally-set sufficient growth standard, is below $\frac{\partial s_{it}(\theta_{it}, \alpha(g_{it}=j+1))}{\partial \theta}$, then the student's score is expected to grow enough to pass the standard if he is promoted from grade 1 to grade 2. If λ is above $\frac{\partial s_{it}(\theta_{it}, \alpha(g_{it}=j+1))}{\partial \theta}$, the student's score growth is expected to satisfy the standard if he is retained, but not if he is promoted to grade $j+1$. To maximize the sufficient score growth standard only, administrators should set δ_j such that students with $\frac{\partial s_{it}(\theta_{it}, \alpha(g_{it}=j+1))}{\partial \theta} < \lambda$ are retained; this occurs for students with relatively low θ_{it}. As a result, students with relatively low scores are those more likely to satisfy the sufficient growth standard when retained.

To gain some insight into the retention decision under different accountability systems, I perform a Monte Carlo exercise, in which I perform 10,000 simulations of the model. Each iteration simulates one school with three grades for ten years, with 100 new students entering the first grade each year. The parameter assumptions I employ are given in Table 1. I define the factor loading parameter $\alpha_{it} = \frac{1}{\sqrt{g_{it}}}$, which is also the functional form used in Figure 2. I assume that measurement is relatively noisy while the noise in skill accumulation is relatively small: $\epsilon_{it} \sim$ N(0,20). and $\eta_{it} \sim$ N(0,10). I also assume $\theta_{i0} \sim$ N(50,25), and that raw ability grows by 35 points per year in expectation, and thus assign $\gamma_0 = 35$. This means that the average student entering the

school will score a 50 on the first grade exam in expectation, and if promoted each year, she will score a 60 on the second grade exam and a 69 on the third grade exam.[2] Finally, I assume that the state passing standard, π, is equal to 50, such that the average student would be expected to pass the exam each year without needing to be retained. I assume that the state sufficient growth standard, λ, is equal to 10, about the rate at which the average student's scores are expected to naturally increase in the absence of retention. An administrator of a school with students scores centered substantially below or above the passing standard, or one whose student population's scores are expected to grow substantially slower or faster than the sufficient growth standard, could behave differently than the one I simulate.

The simulation shows that an administrator will use different retention strategies to maximize school-wide passing rate than he will to maximize both passing rate and sufficient growth rate. Figure 3 summarizes the results of the simulation. To maximize both passing and sufficient score growth rate, an administrator should set a higher δ_1 on average than he would to maximize passing rate only, and a much lower δ_3. δ_2 should be set similarly as under a passing rate-based system. The optimal δ_1, δ_2, and δ_3 values result in 2.925 percentage-point higher retention rates in the first grade on average (a 37.5% increase), 0.201 percentage-point lower retention rates in grade 2 (a 3% decrease), and 6.485 percentage-point lower retention rates in the third grade on average (a 77% decrease). The average retention rate in the school decreases in the presence of growth-based accountability. These results are not surprising given the mechanisms shown in Figure 2. Students in grade 3 that score near the passing threshold π are worth retaining under passing rate-based accountability. The same students are promoted under growth-based accountability, due to the fact that higher-scoring students are likely to exhibit relatively low score growth. I focus on these predictions of the model - that adoption of growth-based accountability leads to lower overall retention rates, lower retention rates in the final grade offered by a school, and higher retention rates in early grades offered by a school - in my empirical analysis.

4 Data

I use a novel source of administrative data on retention rates at the school-grade level from the 2004-05 to the 2025-26 school years, though no state covers the entire study period. The panel was built from public files published by the education agencies of six states: Arkansas, Louisiana, Massachusetts, Michigan, Texas, and Wisconsin.[3] The panel includes retention rates and school identifying information for every school in each state over this 7-year span. Some states publish subgroup-specific retention rates; my primary sample keeps only all-student retention rates for grades K-8, which corresponds to 1,090,656 school-grade cells in 19,500 schools.

[2] The maximum achievable score on the exam is 100.

[3] A seventh, Rhode Island, is collected but not yet incorporated due to inconsistencies in the Rhode Island Department of Education's retention reporting.

Retention reporting is, as one might expect, not uniform across the six states in my sample. Arkansas, Michigan, and Wisconsin publish yearly grade-level counts of retained students, which I combine with yearly grade-level counts of enrolled students to compute the rate.[4] Louisiana, Texas, and Massachusetts publish rates which I use directly. The denominators of the rates may correspond to different objects in different states. In Arkansas and Michigan, the denominator corresponds to fall enrollment, which may be higher or lower than enrollment at the time of the retention decision. In Wisconsin, the denominator captures enrollment at the end of the school term. In Texas, the denominator captures students enrolled during the year who return the following year or graduate, with retention observed the following fall. Louisiana and Massachusetts do not publish the definition of the denominator in their reported retention rates. Using Texas' definition relative to a fall headcount mechanically inflates the rate. Each state uses a single definition throughout the panel, so I estimate proportional effects with school-by-grade fixed effects; these absorb any constant difference in scale between definitions.

Students who repeat courses in high school are sometimes defined as retained students among the states in my sample, while retained students in elementary and middle schools repeat entire grades. High school students that repeat a course may not separated from their cohort in a meaningful way, and because of the different definition, the reported retention rates of high schools differ dramatically from those of middle schools. As such, I restrict my analysis to grades K-8. I do not drop schools that serve in-sample grades that also serve grades 9-12; I only drop those grades.

For each school and year, I measure the highest grade the school served using the state's own files. I consider a grade "served" if it reports non-zero enrollment, retention count, or retention rate. I consider a given grade the school's terminal grade if it equals that year's within-school maximum. Because grade span is measured each year, I pick up terminal grade changes for schools that add or drop a grade over time. On the other hand, I will also pick up a terminal grade change for, say, a small school that happens to have zero enrollment in their terminal grade in a given year. I also use my grade span measurement to classify school types: schools that serve grades 9-12; elementary schools (highest offered grade of 6 or lower); middle schools (lowest offered grade of 5 or higher); K-8 schools (lowest offered grade less than or equal to 2, highest offered grade 8); and other schools (0.6% of cells).

I combine this data with data on the components of states' accountability systems over the same period. I constructed this dataset from 349 primary documents: statutes and administrative rules; state accountability manuals; No Child Left Behind (NCLB) accountability manuals; Every Student Succeeds Act waiver requests; and technical reports and board materials. I code the components, weights, and student populations of interest of each school rating system I observe, as well as what states and years they applied to. I derive my treatment indicators from this coding. Table 2 shows the timing of adoption of growth-based accountability for each state in my sample.

Different states define measures that rate schools on individual student growth differently; some states' definitions change over time. Most states that include a mea-

[4]Michigan began reporting rates in 2013-14, and I use those directly.

sure of individual student growth in school ratings measure the percent of students in each school whose test score increased by a sufficient amount from one year to the next (as defined by the state agency). Some states use the difference between a student's score and a comparable group of students' scores from the previous year, to attempt to average out idiosyncrasies in test performance. Some use projection-based measures of growth, some use "gain indices". All definitions emphasize year-to-year score increases. For the purposes of this paper, I define a state-year as "treated" if its accountabilty system contains components that measure growth for individual students (not cohorts, e.g. NCLB safe-harbor provisions) and components that apply for all students in a school (not just below-proficiency students).

I also code each state's test-score promotion requirements by grade and year since a rule that conditions promotion on a test score constrains the retention decision. In the estimation window, only Texas has one: its Student Success Initiative required a passing score for promotion in grades 3, 5 and 8. The grade 3 requirement was in force through 2008-09, grade 5 through 2010-11, and grade 8 from 2007-08 through 2010-11; grades 5 and 8 were suspended in 2011-12. Neither Arkansas's nor Massachusetts's rules contain such a requirement. Arkansas retains a student who fails the test only if the student also does not take part in remediation. I do not code this as a promotion gate.

I use this multi-state panel to analyze the effect that growth-based accountability systems have on public school retention practices. By using a panel dataset, I am able to exploit the staggered adoption of growth-based accountability across states, comparing the difference in retention practices in states that adopt growth-based accountability to the difference in those that don't, across the time period spanned by my data.

Table 3 describes the grade-level estimation sample: 290,079 school-grade-years in 10,614 schools. The mean retention rate is 2.75%. It is highest in grades K-2 (4.03%) and lowest in grades 4-5 (1.45%), and 43.7% of cells record no retained student. Retention is lower in Massachusetts (1.52%) than in Arkansas (2.5%) or Texas (2.99%). 19% of observations are terminal grades.

5 Empirical analysis

5.1 Measuring the Impact of Growth-based Accountability on Retention Practices

I exploit the gradual adoption of growth-based accountability systems over time across states to study its effect on retention practices. The empirical strategy I use in this paper builds on a large literature in applied economics, which uses variation in timing of treatment to estimate a difference-in-differences effect.

My empirical strategy compares the difference in retention rates in states that adopt growth-based accountability systems before and after the switch to the difference for those that don't over the same time period. I use an event-study design, which al-

lows for analysis even with few untreated units (Abraham and Sun, 2018). In addition, the event-study design allows for varying treatment effects over time; using an event study framework will allow an examination of a potential adjustment period by administrators. The difference-in-differences framework is likely to produce biased estimates in the presence of treatment effects that vary over time (Goodman-Bacon, 2018; Imai and Kim, 2020; Sun and Abraham, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021). As such I follow Sun and Abraham (2021): The latest adopters serve as controls, and the window ends in the year before they adopt; a state enters into estimation only if it is observed before its own adoption; a state never observed at or after its own adoption serves as a not-yet-treated control; a treated cohort with no observation in the year before it adopts in dropped, since its effects are not identified without that reference year.

My empirical approach relies on the exogeneity of the adoption of growth-based accountability systems to school-level decision-makers. While the timing of a state's adoption of a new accountability system and the content of that system are not likely to be random, it is likely to be exogenous to school administrators and their behavior. Accountability systems are adopted by state legislators and designed by state education agencies; in addition, retention practices are not mentioned in any of the accountability systems analyzed in this paper.

Because the effect is predicted to be most salient in a school's terminal grade, I employ a triple-differences design of the following form:

$$\mathbb{E}\left[r_{cgt}\right] = \exp\Big(\sum_{k\neq -1} \beta_k D^k_{st} L_{cgt} + \lambda L_{cgt} + \alpha_{cg} + \gamma_{gt} + \delta_{st}\Big), \tag{11}$$

where r_{cgt} represents the retention rate (a proportion) in grade g of school c in year t; L_{cgt} is equal to one if g is school c's terminal grade in year t; D^k_{st} is equal to one if state s is treated and $t - E_s = k$ (where E_s represents state s's adoption year); α_{cg} are school-by-grade effects, γ_{gt} grade-by-year effects and δ_{st} state-by-year effects. The state-by-year effects absorb the event-time main effects D^k_{st}, along with any shock that moves all of a state's grades proportionally in a given year. A coefficient β corresponds to a $100(e^{\beta} - 1)$ percent change in the last-grade rate relative to the school's other grades. In all specifications, standard errors are clustered at the school level. Though treatment is assigned at the state level, with only three clusters, clustering by state is infeasible. School-level clustering still ignores a state-year shock specific to the difference between the terminal and other grads, and to that extent standard errors will understate uncertainty.

β_k, then, is identified via variation between schools with a given grade that is terminal and non-terminal (e.g. grade 5 in a K-5 school against grade 5 in a K-8 school within Arkansas's state-year and net of that grade's year effect) and from within schools whose grade spans change; notably the latter is the only variation that identifies λ. The identifying assumption is that, absent adoption, Arkansas's within-grade gap between terminal and other grades would have stayed at its 2006-07 level, after grade-by-year and state-by-year effects.

I estimate equation (11) by Poisson pseudo-maximum likelihood (PPML). In 43.7% of cells in my sample, no student is retained; as a result the log of the retention rate is undefined for a large portion of the sample. PPML estimates a conditional mean of exponential form consistently without a distributional assumption (Wooldridge, 1999; Santos Silva and Tenryro, 2006). Terminal grades tend to have low rates overall (1.1% in Arkansas before adoption vs. 3.1% in all other grades). A model in levels gives a given proportional change at the last grade little weight. Finally, if a state's published retention rate equals the true rate times a school-grade specific constant, the log of that constant will be absorbed by α_{cg} in the above, leaving β_k unaffected.

5.2 Results

Table 4 presents the results of my baseline empirical analyses. Columns (1) through (8) report the results of estimating equation (11) with various different fixed effects and controls. Column (3) is the primary specification. The results show a consistent negative terminal-grade effect. The average post-adoption coefficient in the terminal grade is -0.341 (standard error of 0.078). This corresponds with a 29% reduction in terminal-grade retention. The OLS estimated coefficient has the same sign and is less precise: -0.199 with standard error of 0.115. Column (7) gives the results of estimating column (3) while controlling for promotion gates. Column (8) gives the results of estimating column (7) with controls for promotion gate-terminal grade interaction effects; these additions do not seem to meaningfully change the estimate. Figure 4 plots the coefficients of column (3) over time.

Column (3) of Table 4 presents the results of estimating equation (12). The results suggest negative effects of a similar magnitude to the overall effect found in column (1) in each post-treatment period; perhaps because the impact of promoting a student out of school is certain, the impact of the policy was more immediate in the terminal grades offered by schools. Students in terminal grades are clearly held back at a much lower rate under both growth- and level-based accountability systems; however, they are held back at an even more reduced rate after a change to a growth-based system.

6 Conclusion

This paper provides evidence that public school administrators use retention as a tool in optimizing school ratings. Administrators respond to the adoption of accountability criteria which rate schools based on within-student test score growth by changing the rate at which they retain students. My estimates, based on a 7-state sample, suggest that schools retain 18% fewer students after operating under a growth-based system for at least 6 years. I find that the effect is more pronounced in the final grade offered by a school, where promotion of a student evicts them from the pool of students whose test scores contribute to the school's rating. I provide evidence that the effect of growth-based accountability on retention is delayed, indicating a period of adjustment by administrators, though I cannot rule out the possibility of effects as early as

3 years after the implementation of growth-based accountability. I find little evidence of effect heterogeneity by gender or ethnicity. Overall, administrators choose to retain less when student test score growth affects their schools' ratings, particularly when they are able to remove the student from the test-taking pool via promotion. This result is consistent with strategic retention, though my empirical analysis does not allow me to rule out other explanations.

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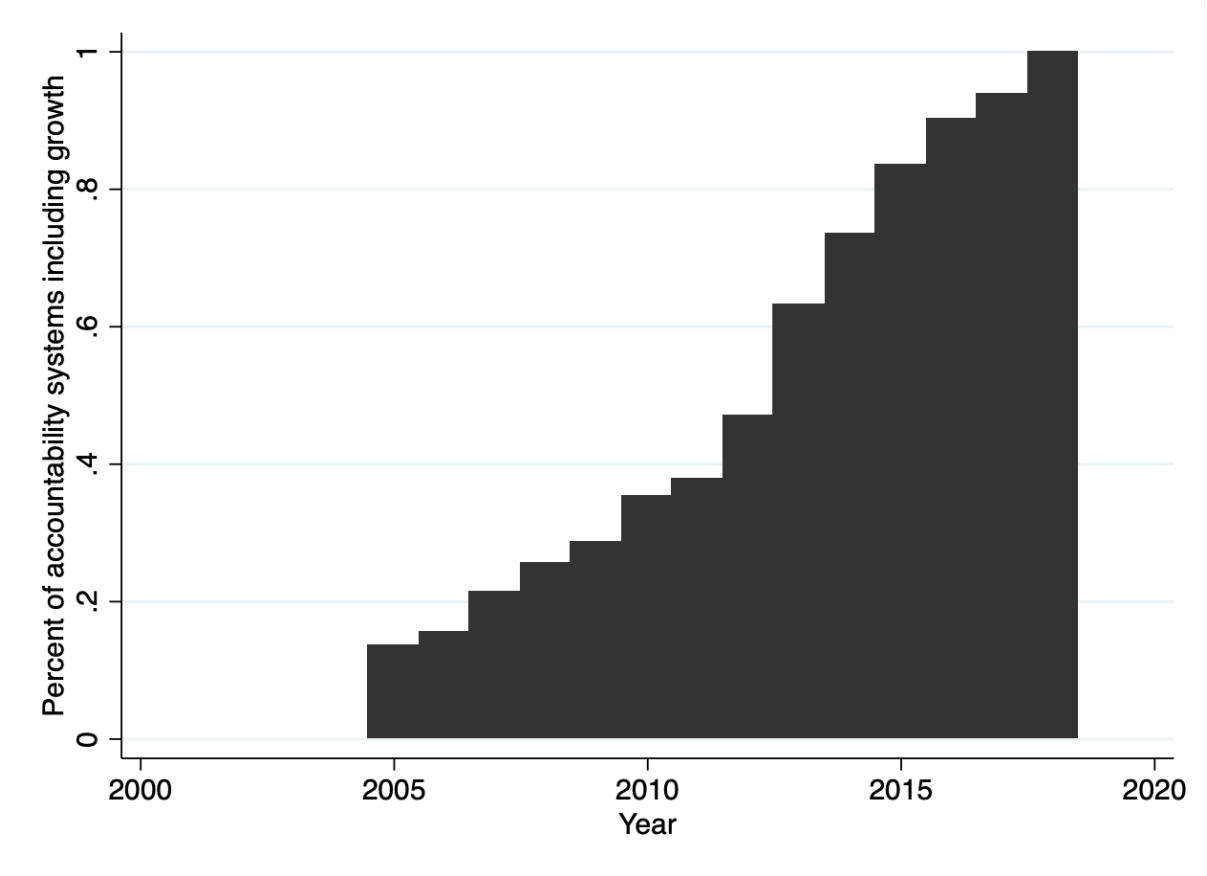
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7 Tables & Figures

Figure 1: Presence of growth in accountability systems over time



Notes: Includes all 50 states and Washington, D.C.
Sources: Education Week Education Counts Research Center; individual states' education agencies.

Figure 2: Level and year-to-year change in student score are affected by grade level

Figure 2a: Student scores for two different grades

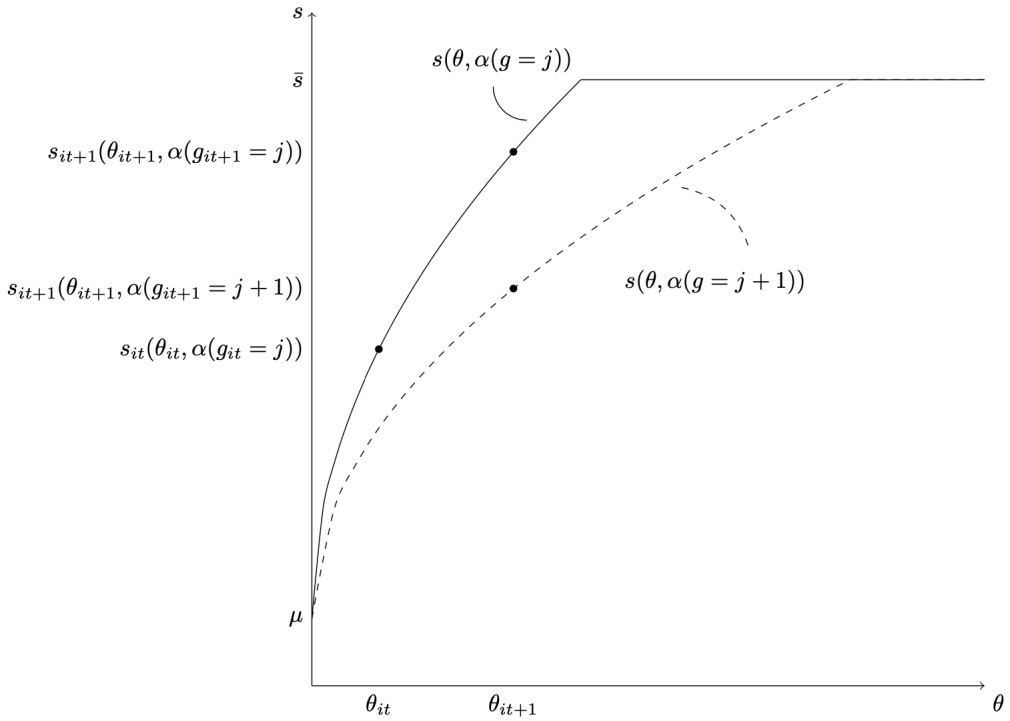


Figure 2b: Rate of change in student scores for two different grades

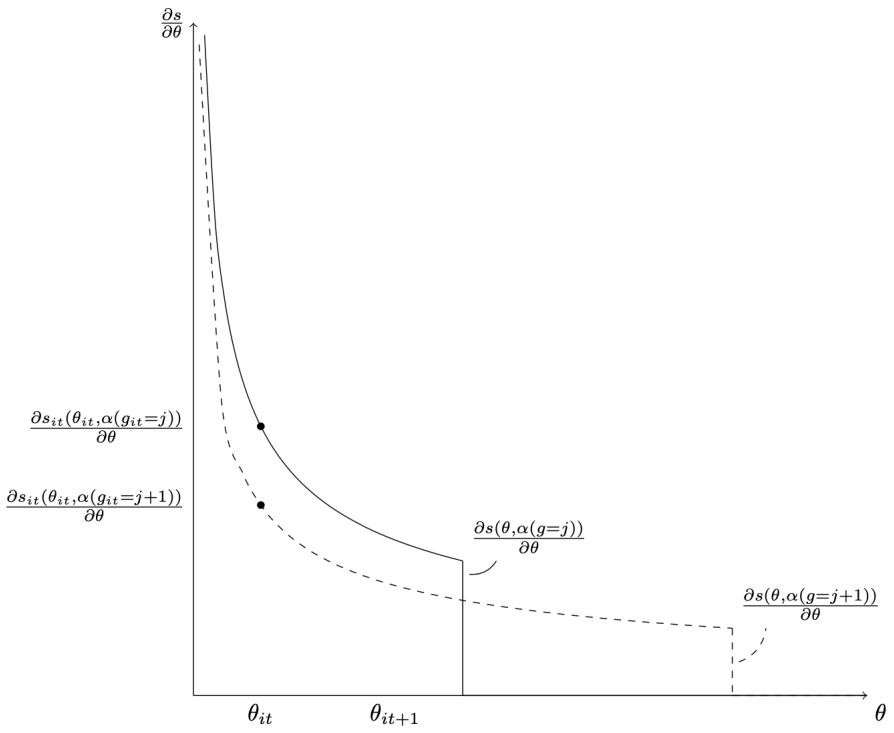


Figure 3: Passing and growth rate-maximizing retention rates

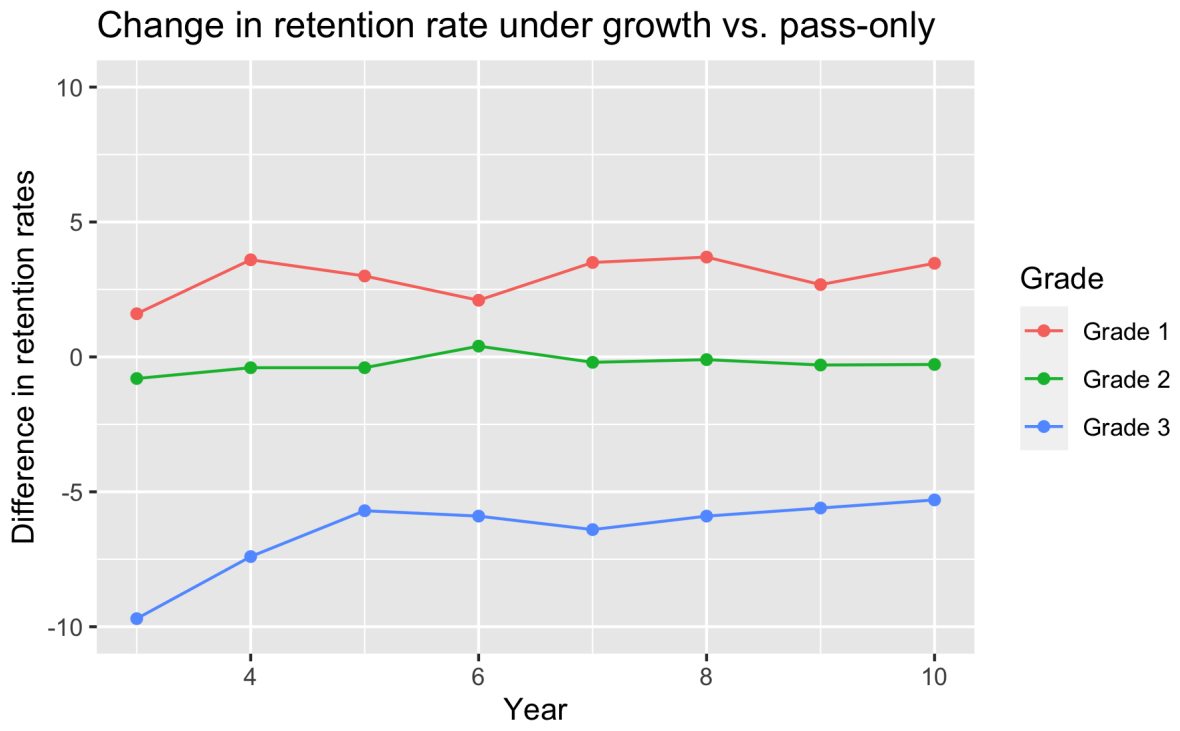


Figure 4: Terminal grade effects over time

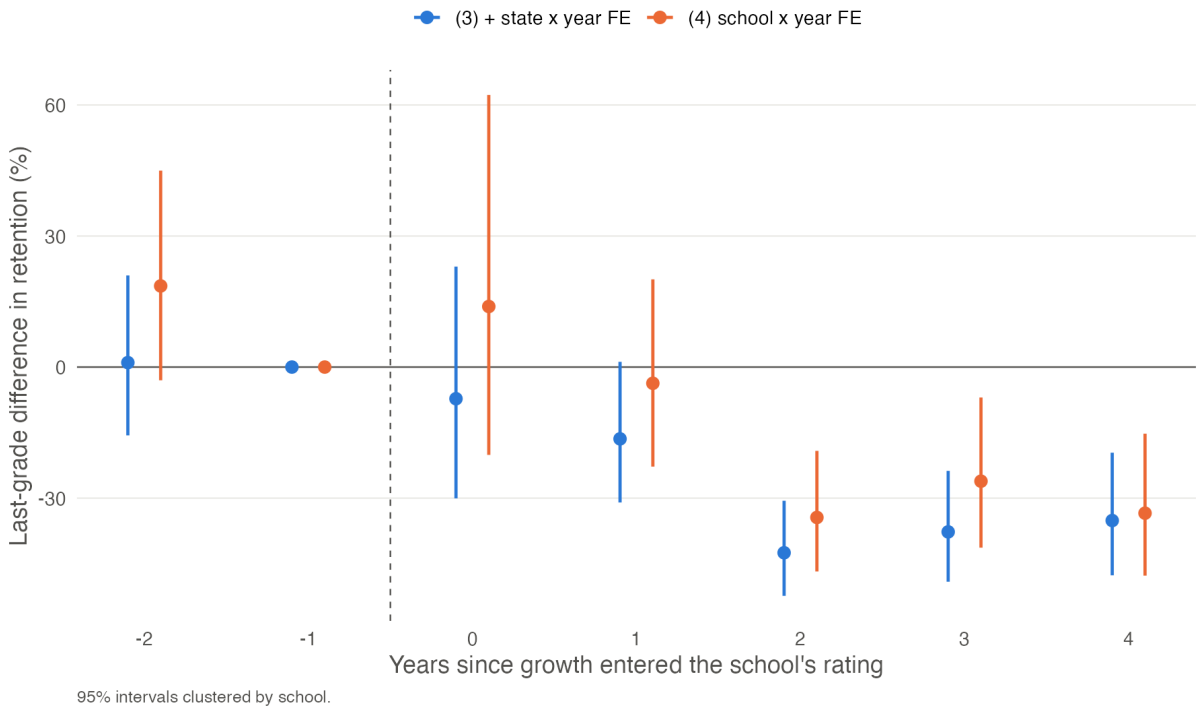


Table 1: Parameter values for simulation

	Mean	S.D.
μ	0	0
ϵ_{it}	0	20
γ_0	35	0
γ_1	1	0
θ_{i0}	50	25[1]
η_{it}	0	10
π	50	0
λ	10	0
Number of students entering grade 1 each year	100	0
Number of years simulated	10	0
Number of simulations	10000	0

Notes:

[1]: Initial θ_{it} draw for students entering school.

Table 2: Timing of adoption of growth-based accountability

	AR	LA	MA	MI	RI†	TX	WI
2004-05							
2005-06							
2006-07							
2007-08	✓						
2008-09	✓			✓		✓	
2009-10	✓			✓		✓	
2010-11	✓			✓			
2011-12	✓		✓	✓	✓†		
2012-13	✓	✓	✓	✓	✓†	✓	✓
2013-14	✓	✓	✓	✓	✓†	✓	✓
2014-15	✓	✓	✓	✓	✓†	✓	
2015-16		✓	✓	✓	✓†	✓	✓
2016-17		✓	–	✓	✓†	✓	✓
2017-18	✓	✓	✓	✓	✓†	✓	✓
2018-19	✓	✓	✓	✓	✓†	✓	✓
2019-20	–	✓	–	n.c.	✓†	✓	–
2020-21	–	✓	–	n.c.	✓†	–	✓
2021-22	✓	n.c.	✓	n.c.	✓†	✓	✓
2022-23	✓	n.c.	✓	n.c.	✓†	✓	✓
2023-24	✓	n.c.	✓	n.c.	✓†	✓	✓

Notes: ✓: individual student growth in the school's rating in either the federal or the state channel, known by the spring retention decision (decision basis). –: no rating issued in any channel. ?: no document in hand. n.c.: outside the state's coded years. Texas: standard campuses. †Rhode Island is not yet coded from documents; its column reproduces the earlier coding (from 2011-12), which EdCounts contradicts.

Table 3: Descriptive statistics: the baseline estimation samples (definition B)

Panel A. School-grade sample, 2006-2012				
	AR	TX	MA	All
Role	Treated	Control	Control	
Adopts growth (definition B)	2007-08	2012-13	2011-12	
First year	2005-06	2005-06	2005-06	2005-06
Last year	2011-12	2011-12	2010-11	2011-12
Schools	1,107	7,813	1,694	10,614
School-grade-years	27,882	223,638	38,559	290,079
Retention rate (%), mean	2.50	2.99	1.52	2.75
(standard deviation)	[4.86]	[6.57]	[4.06]	[6.17]
Mean before the first adoption	2.73	3.52	1.66	3.15
No pupil retained (% of cells)	50.2	40.0	60.6	43.7
Retention rate (%), grades K-2	4.45	4.19	2.53	4.03
Retention rate (%), grade 3	1.29	2.47	1.22	2.17
Retention rate (%), grades 4-5	0.81	1.68	0.70	1.45
Retention rate (%), grades 6-8	1.81	2.68	1.41	2.41
School's last grade (% of cells)	18.4	18.6	22.0	19.0
Elementary school (% of cells)	77.5	73.0	68.8	72.9
Middle school (% of cells)	10.1	12.3	14.4	12.4
K-8 school (% of cells)	1.4	3.3	12.2	4.3
School with grades 9-12 (% of cells)	10.2	10.9	3.5	9.9
Other school type (% of cells)	0.8	0.5	1.0	0.6
Promotion gate in force (% of cells)	0.0	22.0	0.0	16.9
Exact retention count (% of cells)	100.0	42.8	0.0	42.6
Denominator	Fall count	Stayers	None	

Notes: Retention rates in percent; means and standard deviations are unweighted over cells. School type comes from the grades a school serves, except in Wisconsin, where it is the state's own elementary/middle grouping. Before 2013-14 Michigan publishes no row for a school that retained nobody; its zeros here are filled from the agency's district totals, where a district's total equals the sum of its schools. Elsewhere such a school remains absent. .

Table 4: Triple difference: the school's last grade (PPML, log points): Baseline: definition B

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Year -2 × last grade	-0.333	0.011	0.010	0.170	0.025	-0.043	0.010	0.014
	(0.074)	(0.092)	(0.092)	(0.103)	(0.102)	(0.092)	(0.092)	(0.092)
Year 0 × last grade	-0.365	-0.076	-0.075	0.130	0.073	-0.087	-0.060	-0.055
	(0.139)	(0.144)	(0.144)	(0.181)	(0.172)	(0.146)	(0.145)	(0.145)
Year 1 × last grade	-0.519	-0.180	-0.180	-0.038	-0.017	-0.148	-0.163	-0.159
	(0.070)	(0.098)	(0.098)	(0.113)	(0.114)	(0.098)	(0.098)	(0.098)
Year 2 × last grade	-0.908	-0.553	-0.554	-0.422	-0.409	-0.524	-0.536	-0.532
	(0.078)	(0.096)	(0.096)	(0.107)	(0.110)	(0.097)	(0.095)	(0.096)
Year 3 × last grade	-0.820	-0.474	-0.474	-0.303	-0.230	-0.456	-0.455	-0.451
	(0.083)	(0.103)	(0.103)	(0.118)	(0.119)	(0.104)	(0.102)	(0.103)
Year 4 × last grade	-0.905	-0.433	-0.433	-0.407	-0.242	-0.414	-0.439	-0.436
	(0.090)	(0.110)	(0.110)	(0.123)	(0.127)	(0.110)	(0.110)	(0.110)
Average post-adoption × last grade	-0.702	-0.342	-0.341	-0.201	-0.160	-0.324	-0.329	-0.325
	(0.056)	(0.078)	(0.078)	(0.093)	(0.091)	(0.079)	(0.078)	(0.078)
Last grade offered	-0.241	-0.074	-0.073	-0.183	-0.139	-0.100	-0.073	-0.086
	(0.026)	(0.071)	(0.071)	(0.054)	(0.052)	(0.065)	(0.071)	(0.075)
Promotion gate in force							0.102	0.096
							(0.041)	(0.043)
Promotion gate × last grade								0.032
								(0.059)
Year -2	-0.039	-0.070						
	(0.036)	(0.037)						
Year 0	0.034	0.004						
	(0.038)	(0.033)						
Year 1	0.158	0.106						
	(0.044)	(0.044)						
Year 2	0.170	0.099						
	(0.040)	(0.038)						
Year 3	0.186	0.122						
	(0.042)	(0.038)						
Year 4	0.141	0.054						
	(0.041)	(0.040)						
Observations	286,478	254,716	254,716	236,332	236,332	254,716	254,716	254,716
Schools	9,995	9,677	9,677	8,906	8,906	9,677	9,677	9,677
Campus FE	✓							
Grade FE	✓							
Year FE	✓							
School × grade FE		✓	✓	✓	✓	✓	✓	✓
Grade × year FE		✓	✓		✓	✓	✓	✓
State × year FE			✓			✓	✓	✓
School × year FE				✓	✓			
School type × year FE						✓		
Promotion gate control							✓	✓
Gate × last grade								✓

Notes: Standard errors, clustered at the school level, are given in parentheses.Column (7) adds an indicator for a state test-score promotion gate in force on the grade that year (Texas grades 3, 5 and 8 while in force; Arkansas and Massachusetts have none). Column (8) also lets the gate differ where the gated grade is the school's last. A coefficient b is a $100(e^b - 1)\%$ change.